

AUTO DATA STORYTELLER REPORT

Executive Summary

****Executive Summary**** Our newly developed predictive model demonstrates strong and reliable performance, accurately identifying outcomes in over 83% of cases. This high level of precision ensures that the insights derived are trustworthy and can confidently inform critical business decisions and strategic planning. Analysis of the model reveals that an individual's gender (specifically being male), the fare paid, and age are the most significant factors influencing the observed outcomes. Understanding the profound impact of these demographic and transactional variables is crucial for developing targeted strategies. We recommend continuous monitoring of these key drivers, as changes in these areas are likely to have the most substantial impact on future outcomes. Proactively tracking them will enable us to anticipate trends, mitigate potential risks, and optimize our business strategies effectively based on reliable predictions.

Dataset Overview

Problem Type: classification

Best Model: Random Forest

Data Quality

- Dataset contains 891 rows & 12 columns
- 866 missing values were identified.
- Column Cabin has the highest missing values (687 values)
- No strong relationships were identified.

Model Findings

CLASSIFICATION MODEL PERFORMANCE

- Random Forest was selected as the best performing model based on F1.
- Random Forest achieved the highest F1 score of 0.83.
- The model correctly classified approximately 83.2% of records.
- The model demonstrates strong classification performance.
- Random Forest outperformed Logistic Regression, Decision Tree based on F1.

Key Drivers

MODEL DRIVERS

- Sex_male was the most influential feature.
- Fare was among the strongest predictors.
- Age contributed significantly to model decisions.

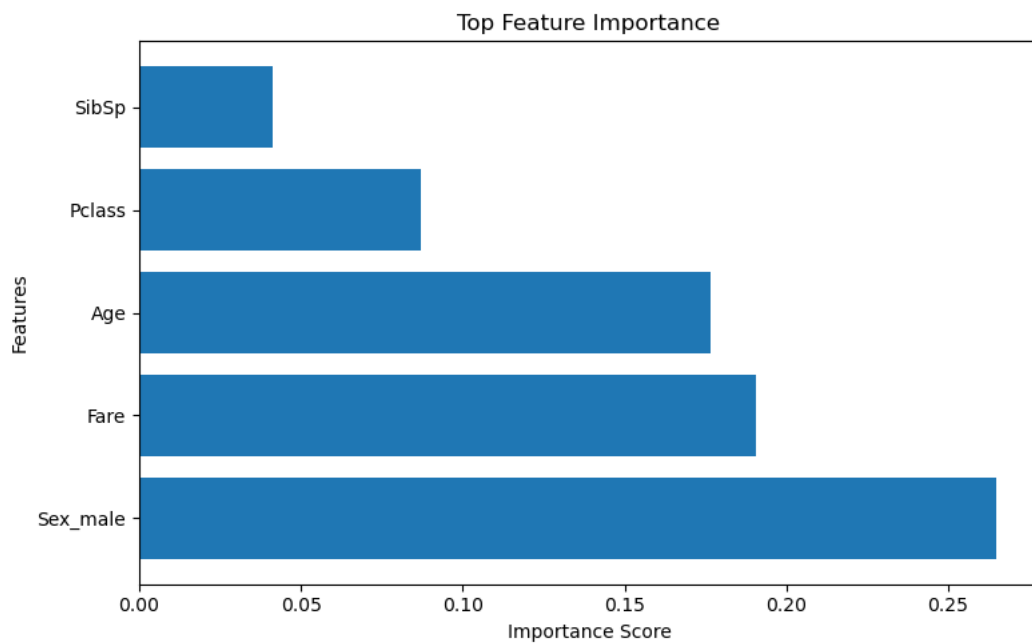
Recommendations

RECOMMENDATIONS

- Sex_male appears to be the strongest driver of predictions and should be monitored closely.

- The model demonstrates strong predictive performance, making the insights reasonably reliable.
- The most influential features should be tracked regularly, as changes in these variables are likely to have the largest effect on outcomes.

Feature Importance



Missing Values

